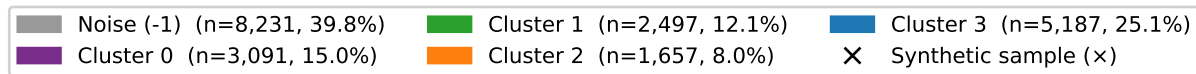
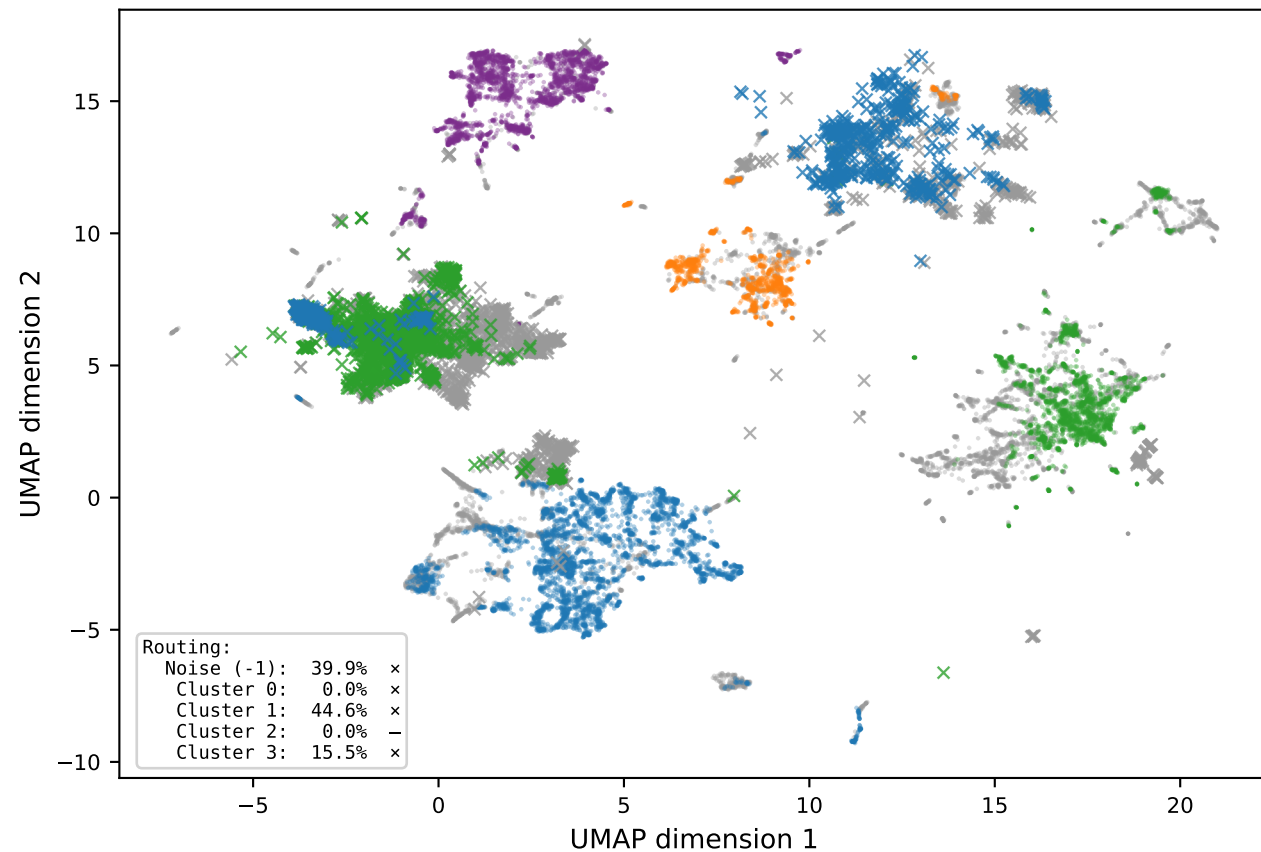


Fraud cluster structure (UMAP) and synthetic allocation: CTGAN vs TVAE
Real fraud n=20,663 · colours = HDBSCAN cluster · crosses = synthetic samples



CTGAN (150 ep) — fold 4
n=6,182 synthetic



TVAE (300 ep) — fold 4
n=5,597 synthetic

